

Active SolvAI: learning what to measure in alchemical hydration calculations

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Abstract

Accurate hydration free energies require target-specific sampling, whereas structure-only models can fail under chemical shift. Active SolvAI asks whether sparse response observations from the query molecule can update a transferable SolvAI prior and recover either the same-Hamiltonian alchemical response or the experimental endpoint at lower measured cost. This manuscript is a living preregistered report: quantitative claims are inserted only after their generating analysis has been frozen and completed.

1 Introduction

SolvAI established that structure-predicted solvent responses add information beyond matched molecular features and experimental labels. It did not establish that high-fidelity response can be recovered from structure alone, nor that uncertainty can guide a simulation. We therefore test a different proposition: actual sparse observations may reveal molecule-specific departures from the structure prior.

2 Results

2.1 Actual-observation gate

PHASE1_RESULT_PENDING. The primary estimand is the paired held-out endpoint-error change obtained from actual-minus-prior response residuals relative to frozen SolvAI. Same-Hamiltonian reconstruction and experimental prediction are reported separately.

2.2 Retrospective replay

REPLAY_RESULT_PENDING. This section will be populated only if the preregistered Phase 1 go criterion is met.

2.3 Prospective sentinels

PILOT_RESULT_PENDING. This section will be populated only after a replay policy and its comparators are frozen.

3 Discussion

The study is designed to support a positive, neutral or negative conclusion without changing its scientific question. Any benefit for same-Hamiltonian reconstruction will not by itself be described

as improved agreement with experiment; any endpoint benefit without curve reconstruction will be described as an empirical residual diagnostic rather than successful quadrature.

4 Methods

The parent SolvAI artifact, cohorts, splits and response campaigns are versioned by SHA-256. All preprocessing, calibration and model selection occur inside molecule-held-out training data. Destructive controls preserve response distributions while shuffling molecule alignment. Every attempted run, including failures, is included in the compute ledger.